

MULTI-ANATOMY ADAPTATION AND CONFIGURATION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device is configured to operate across multiple anatomical environments while maintaining a common sensing architecture. The device may be adapted for use within vaginal, anal, or other luminal anatomical structures without requiring fundamental redesign of sensing or computational systems.

Adaptation to different anatomical environments may be achieved through variation of insertable length, diameter, flexibility, surface compliance, or distal geometry. Devices may be produced in multiple sizes or adjustable configurations while maintaining standardized sensing operation and data interpretation frameworks.

In certain embodiments, interchangeable tips, sleeves, or modular insertable sections may be provided to accommodate anatomical variation or user preference. Modular adaptation may allow a shared handle and electronics platform to support multiple anatomical configurations through replaceable mechanical components.

The device may further adapt sensing behavior based on detected anatomy or selected operating mode. Software configuration profiles may adjust inflation limits, sensing thresholds, stabilization requirements, or guidance workflows according to the intended measurement environment.

Distance sensing algorithms may compensate for differences in lumen curvature, reflectivity, acoustic coupling, or tissue compliance encountered across anatomical environments. Computational normalization may ensure that derived outputs remain comparable despite variation in anatomy or device configuration.

In embodiments incorporating inflatable compliance structures, expansion limits may be dynamically restricted based on selected anatomy to maintain safe operating conditions. Inflation control logic may modify rate, maximum pressure, or allowable expansion ranges depending on configuration mode.

The device may also support anatomy-agnostic operation in which sensing data is interpreted independently of anatomical labeling. In such implementations, computational analysis derives structural and compliance characteristics without reliance on predefined anatomical assumptions, thereby preserving flexibility for future applications.

Multi-anatomy adaptation embodiments may include ergonomic variations such as handle orientation changes, insertion angle adjustments, or positioning aids designed to improve usability across different measurement contexts.

Accordingly, the disclosed architecture enables a unified intraluminal sensing platform capable of serving multiple anatomical applications while maintaining common calibration standards, shared computational infrastructure, and consistent user profiling methodologies.